Nest-Site Temperature and Hatchling Sex Ratios in Green Sea Turtles: A Pilot Monitoring Study

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Abstract

Sea turtles show temperature-dependent sex determination: warmer incubation produces more females, and cooler incubation produces more males. As coastal temperatures rise, this mechanism may skew hatchling sex ratios and threaten population stability. In this pilot study we monitored 62 green sea turtle (Chelonia mydas) nests across three beaches during one nesting season. Temperature loggers recorded sand conditions at nest depth, and hatchling sex was inferred from incubation temperature during the middle third of development. Nests on the unshaded beach incubated on average 2.1 degrees C warmer than nests on the vegetated beach, and produced an estimated 91% female hatchlings compared with 63%. Simple shading and the relocation of threatened nests to cooler zones reduced mean incubation temperature by 1.4 degrees C. We conclude that low-cost management actions can partially offset warming, although longer monitoring is needed.

1. Introduction

Green sea turtles return to the beaches where they hatched to lay their eggs. Because the sex of the embryos is determined by the temperature of the sand rather than by sex chromosomes, small changes in climate can have large demographic effects. Previous work on several turtle populations has reported strongly female-biased hatchling cohorts in warm regions, which raises concern about the future availability of adult males.

The aim of this pilot study was to describe how nest-site conditions on three beaches relate to incubation temperature and estimated sex ratio, and to test whether inexpensive interventions can lower nest temperatures during the hottest weeks of the season.

2. Methods

Study sites. We selected three beaches with different vegetation cover: an open, unshaded beach, a partially shaded beach, and a beach backed by dense coastal forest. Nest monitoring ran for 16 weeks during the nesting season.

Temperature recording. Each nest received a miniature data logger buried at the center of the egg chamber, recording sand temperature every 30 minutes. Mean temperature during the middle third of incubation was used as the predictor of sex, with a pivotal temperature of 29 degrees C assumed for this population.

Interventions. Fifteen nests on the unshaded beach were assigned to one of two treatments: shade cloth placed over the nest, or relocation to a cooler zone nearer the vegetation line. Fifteen untreated nests served as controls.

3. Results

Across all sites, incubation temperature ranged from 27.4 to 33.1 degrees C. Nests on the unshaded beach averaged 31.2 degrees C, nests on the partially shaded beach 30.0 degrees C, and nests on

the forested beach 29.1 degrees C. The estimated proportion of female hatchlings was 91%, 79% and 63%, respectively.

Both interventions lowered temperature. Shade cloth reduced mean incubation temperature by 1.2 degrees C and relocation by 1.6 degrees C relative to controls. Hatching success did not differ between treatments and controls (82% and 80%).

4. Discussion

The results show that vegetation cover is a strong predictor of nest temperature and that heavily female-biased cohorts are already likely on open beaches. Shading and relocation are cheap, practical measures that local conservation teams can apply, and they did not reduce hatching success in this sample.

Relocation brings its own risks, including disturbance of the eggs and the possibility of moving nests into areas that flood. We therefore recommend prioritizing nests that are most exposed and using shade where possible.

5. Limitations

The study covers a single season and three beaches, with a modest number of nests. Sex ratios were estimated from temperature rather than confirmed by direct examination of the hatchlings. The pivotal temperature was assumed, not measured, for this population. Rainfall and nest depth were not controlled. Results should therefore be treated as preliminary.

6. Conclusion

Nest-site temperature strongly influences the estimated sex ratio of green sea turtle hatchlings, and simple management actions can lower temperatures by more than one degree. Multi-season monitoring with direct sex confirmation is the next step.

Illustrative test document written for a software demonstration. Authors, institutions and all data are fictional.